Extremal degree multiplicities for modified total irregularity of chemical graphs

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Abstract

For a connected simple graph on n vertices with maximum degree at most four, consider the sum of $|d(u) - d(v)|^{1/n}$ over all unordered pairs of distinct vertices. We determine every degree-multiplicity vector attaining the maximum, for every $n \geq 7$. This proves the classification proposed in Conjecture 10 of Knor, Škrekovski, Filipovski and Dimitrov. The proof first optimizes over integer multiplicities satisfying the handshaking parity condition. A positive quadratic decomposition reduces the problem to two nearest-integer comparisons. We then show that every maximizing vector has a connected simple realization. The argument applies to all $n \geq 7$ and does not depend on finite enumeration.

1 Statement and relation to previous work

A *chemical graph* here means a finite, undirected, connected simple graph of maximum degree at most four, as in [1]. For such a graph G of order $n \geq 7$, define

$$I_n(G) = \sum_{\{u,v\} \in \binom{V(G)}{2}} |d(u) - d(v)|^{1/n}. \tag{1}$$

The sum is over all unordered pairs of distinct vertices, not only edges. Write $a_i = |\{v \in V(G) : d(v) = i\}|$ for $1 \leq i \leq 4$.

In arXiv:2411.01530v1, Knor et al. [1, Theorem 9 and Conjecture 10] proved a classification for a smaller, order-dependent range of positive exponents, and conjectured that the same classification holds for exponent $1/n$. Dimitrov et al. [2, Theorem 2.2] subsequently reported the $n \equiv 0 \pmod 4$ case, within a more general parameter statement. We give a self-contained proof for all four residue classes; no result about the more general exponent c/n is used below. The constant-exponent Conjecture 11 of [1] is a different question.

Theorem 1. *For every integer $n \geq 7$, a chemical graph of order n maximizes* (1) *if and only if its degree-multiplicity vector occurs in Table 1. Each listed vector has a connected simple realization.*

The last row is the vector $(K, K-1, K, K)$ when $n = 4K-1$, as written in [1]. The classification concerns degree multiplicities, not isomorphism types. Any two graphs with the same multiplicities have the same value of (1).

Order	(a_1, a_2, a_3, a_4)
$4k$	(k, k, k, k)
$4k+1$	$(k, k, k, k+1)$
$4k+2$	$(k, k+1, k, k+1)$ or $(k+1, k, k+1, k)$
$4k+3$	$(k+1, k, k+1, k+1)$

Table 1: All maximizing degree multiplicities, with $n \geq 7$.

2 Optimization over integer multiplicities

Every chemical graph gives a vector in the larger set

$$\mathcal{A}_n = \left\{ a \in \mathbb{Z}^4_{\geq 0} : \sum_{i=1}^{4} a_i = n, \quad a_1 + a_3 \equiv 0 \pmod 2 \right\}.$$

The parity condition follows from the handshaking lemma. We first maximize over $\mathcal{A}_n$ without requiring graphicality. Put

$$p = \frac{1}{n}, \qquad T = 2^p, \qquad S = 3^p, \qquad D = 1 + 2T - S.$$

Counting pairs by their degrees gives

$$Q(a) = a_1a_2 + a_2a_3 + a_3a_4 + T(a_1a_3 + a_2a_4) + Sa_1a_4. \tag{2}$$

For every graph with multiplicities a, its index equals $Q(a)$.

Set $s = a_1 + a_4$, $u = a_1 - a_4$, and $v = a_2 - a_3$. The inverse change of variables is

$$a_1 = \frac{s+u}{2}, \quad a_4 = \frac{s-u}{2}, \quad a_2 = \frac{n-s+v}{2}, \quad a_3 = \frac{n-s-v}{2}. \tag{3}$$

Expansion of (2) yields

$$4Q(a) = n^2 + 2nTs - Ds^2 - E(u, v), \tag{4}$$

$$\begin{aligned} E(u, v) &= Su^2 + v^2 + 2(T-1)uv \\ &= (S - T + 1)u^2 + (2 - T)v^2 + (T-1)(u+v)^2. \end{aligned} \tag{5}$$

All three coefficients in (5) are strictly positive: $1 < T < 2$ and $S > T$. The integer and parity conditions become

$$u \equiv s \pmod 2, \qquad v \equiv n - s \pmod 2, \qquad u - v \equiv -n \pmod 4. \tag{6}$$

Nonnegativity additionally requires $0 \leq s \leq n$, $|u| \leq s$, and $|v| \leq n - s$.

2.1 The parity penalty

The congruences give the bounds and equality conditions in Table 2. These are uniform lower bounds; at an arbitrary boundary value of s, their equality conditions need not be feasible.

Here is a verification of every case. If $n \equiv 0 \pmod 4$ and s is even, positivity in (5) gives $E \geq 0$, with equality only at $(0, 0)$. If $n \equiv 0 \pmod 4$ and s is odd, then u, v are odd and $u+v \equiv 2 \pmod 4$. Thus the three squares in (5) are at least $1, 1, 4$, respectively. Equality requires $u = v = \pm 1$.

	s even		s odd	
$n \bmod 4$	Bound on E	Equality (u, v)	Bound on E	Equality (u, v)
0	0	$(0, 0)$	$S + 2T - 1$	$(1, 1),\ (-1, -1)$
1	1	$(0, 1)$	S	$(-1, 0)$
2	4	$(0, 2),\ (0, -2)$	$S + 3 - 2T$	$(1, -1),\ (-1, 1)$
3	1	$(0, -1)$	S	$(1, 0)$

Table 2: Lower bounds for $E(u, v)$ under (6).

If n is odd and s is even, then u is even, v is odd, and $u + v$ is odd. Consequently $E \geq (2 - T) + (T - 1) = 1$. Equality requires $u = 0$, $|v| = 1$, and the last congruence in (6) fixes the sign of v. If n and s are odd, the corresponding bound is $E \geq (S - T + 1) + (T - 1) = S$, with $|u| = 1$, $v = 0$ and the sign of u again fixed by that congruence.

If $n \equiv 2 \pmod 4$ and s is odd, then u, v are odd, giving $E \geq (S-T+1)+(2-T) = S+3-2T$. Equality holds exactly when $u = -v = \pm 1$. Finally, if $n \equiv 2 \pmod 4$ and s is even, then u, v are even, $u - v \equiv 2 \pmod 4$, and $u + v \equiv 2 \pmod 4$. Hence $u^2 + v^2 \geq 4$ and $(u + v)^2 \geq 4$. Since $S - T + 1 > 2 - T$, we obtain

$$E \geq 4(2 - T) + 4(T - 1) = 4,$$

with equality exactly at $u = 0$, $v = \pm 2$.

2.2 The nearest integers and comparison of the two parities

Define

$$H(s) = \frac{n^2 + 2nTs - Ds^2}{4}.$$

Strict concavity of x^p on the positive real axis gives $2T > 1 + S$, so $D > 2$. The vertex of H is

$$s_* = \frac{nT}{D} = \frac{n}{2} + \delta, \qquad \delta = \frac{n(S - 1)}{2D}. \tag{7}$$

The binomial theorem, with $n \geq 2$, gives $(1 + 2/n)^n > 1 + n(2/n) = 3$. Thus

$$n(S - 1) < 2, \qquad 0 < \delta < \frac{1}{2}. \tag{8}$$

For fixed n, let $L(s)$ denote the lower bound from Table 2 for the parity of s. Then $Q(a) \leq U(s) := H(s) - L(s)/4$. On each parity class $L(s)$ is constant; consequently its upper bound $U(s)$ is uniquely maximized at the integer of that parity nearest to s_*. Equation (8) makes these two integers explicit in Table 3.

$n \bmod 4$	Even candidate	Odd candidate	Winning s	Winning U minus other U
0	$n/2$	$n/2 + 1$	$n/2$	$(4T - n(S - 1))/4$
1	$(n - 1)/2$	$(n + 1)/2$	$(n + 1)/2$	$(n - 1)(S - 1)/4$
2	$n/2 + 1$	$n/2$	$n/2$	$(2D - n(S - 1))/4$
3	$(n + 1)/2$	$(n - 1)/2$	$(n + 1)/2$	$(n + 1)(S - 1)/4$

Table 3: The two parity candidates and their strict comparison.

Both candidates have $s \geq 3$ and $n - s \geq 3$ when $n \geq 7$. Every equality pair in the corresponding row of Table 2 therefore satisfies the nonnegativity constraints in (3). Thus both upper bounds being compared are attained in $\mathcal{A}_n$. Substitution into U gives the differences in the last column. Each is strictly positive by $S > 1$, $T > 1$, $D > 2$, and (8).

The winning (s, u, v) are therefore

n	(s, u, v)
$4k$	$(2k, 0, 0)$
$4k+1$	$(2k+1, -1, 0)$
$4k+2$	$(2k+1, 1, -1)$ or $(2k+1, -1, 1)$
$4k+3$	$(2k+2, 0, -1)$.

Using (3) gives exactly Table 1. The strict comparison between parities, uniqueness of the nearest integer within each parity, and the equality conditions for E exclude every other maximizing vector in $\mathcal{A}_n$.

3 Connected simple realizations

It remains to show that each optimal integer vector is realized by a chemical graph. We recall the small-degree graphicality result of Dimitrov et al. [2, Proposition 2.2] and include its proof.

Lemma 2. *If $n \geq 8$ and $d_1, \ldots, d_n \in \{1, 2, 3, 4\}$ have even sum, then they are the degrees of a simple graph.*

Proof. Arrange the sequence in nonincreasing order. By the Erdős–Gallai criterion [3], it suffices to check, for $1 \leq r \leq n$,

$$\sum_{i=1}^{r} d_i \leq r(r-1) + \sum_{i=r+1}^{n} \min(r, d_i).$$

For $r = 1$, the left side is at most 4 and the right side is $n - 1 \geq 7$. For $r = 2$, the respective bounds are 8 and $2 + (n-2) = n \geq 8$. For $r = 3$, they are 12 and $6 + (n-3) = n + 3$, which suffice for $n \geq 9$. If $n = 8$ and the inequality at $r = 3$ fails, integrality forces the left side to be 12 and the right side to be 11. The degree sequence must then be $(4, 4, 4, 1, 1, 1, 1, 1)$, whose sum is odd, a contradiction. For $r = 4$, the left side is at most 16 and the right side at least $12 + (n-4) = n + 8 \geq 16$. For $r \geq 5$, the term $r(r-1)$ alone is at least $4r$, which bounds the left side. □

For completeness, we also prove the connected-realization criterion recorded in [3, (1.6)].

Lemma 3. *A graphical sequence of n positive integers with sum at least $2(n-1)$ has a connected simple realization.*

Proof. Start with any simple realization. If it has $c \geq 2$ components and every component is acyclic, then it has $n - c \leq n - 2$ edges, contrary to the assumed degree sum. Thus a component contains a cycle. Choose an edge uv of such a cycle and an edge xy in another component, the latter existing because all degrees are positive. Replace uv, xy by ux, vy. The new edges did not previously exist and have distinct endpoints, so the graph remains simple and every degree is unchanged. Removing uv leaves its component connected. If xy was a bridge, its two resulting pieces both join the first component through the new edges; otherwise its component remains connected and joins the first. In either case the number of components decreases by one. Repeating this operation gives a connected realization. □

Completion of the proof of Theorem 1. The vectors in Table 1 have even degree sums, respectively

$$10k, \qquad 10k+4, \qquad 10k+6 \text{ or } 10k+4, \qquad 10k+8.$$

Each sum is at least $2(n-1)$ in the stated range $n \geq 7$. For $n \geq 8$, Lemmas 2 and 3 therefore give connected simple realizations with all degrees at most four. For $n = 7$, the required sequence is $(4, 4, 3, 3, 2, 1, 1)$; on vertices $0, \ldots, 6$ the edge set

$$\{02, 03, 04, 05, 12, 13, 14, 16, 23\}$$

is an explicit connected simple realization, where ij denotes $\{i, j\}$. Thus the maximum over $\mathcal{A}_n$ is attained in the original graph class. Since every chemical graph gives a vector in $\mathcal{A}_n$, the classification already obtained proves the theorem. □

4 Reproducible finite checks

The supplied `verify.py` script uses only the Python standard library. It enumerates all 317,826 parity-feasible multiplicity vectors for $7 \leq n \leq 60$. It encloses $2^{1/n}$ and $3^{1/n}$ between rational bounds with denominator 2^{48}, certifying the bounds by integer powers. For every nonmaximizing vector it verifies a strictly positive lower bound on the difference from the stated maximum. The two maximizing vectors when $n \equiv 2 \pmod 4$ have identical integer coefficient triples in the basis $(1, T, S)$, so their equality is exact. The script also constructs and checks all 242 listed maximizing degree sequences for $7 \leq n \leq 200$, using degree-preserving edge switches to obtain connected realizations. Edge lists and numerical certificates are included with the script. These are finite consistency checks, not premises of the proof for arbitrary n.

AI use and verification status. This research was completed with AI assistance. The mathematical argument and reproducible checks are supplied for scrutiny. This version does not claim verification in a proof assistant or external peer review.

References

[1] M. Knor, R. Škrekovski, S. Filipovski, and D. Dimitrov, *Extremizing antiregular graphs by modifying total σ-irregularity*, Applied Mathematics and Computation **490** (2025), 129199. doi:10.1016/j.amc.2024.129199. Preprint arXiv:2411.01530v1 (2024), especially Theorem 9 and Conjecture 10.

[2] D. Dimitrov, S. Filipovski, M. Knor, and R. Škrekovski, *Modified Total Sigma-Irregularity in Chemical Graphs*, Discrete Mathematics Letters **18** (2026), 1–7. doi:10.47443/dml.2026.041.

[3] P. Erdős and T. Gallai, *Gráfok előírt fokú pontokkal*, Matematikai Lapok **11** (1960), 264–274 (in Hungarian). Full-text archive.